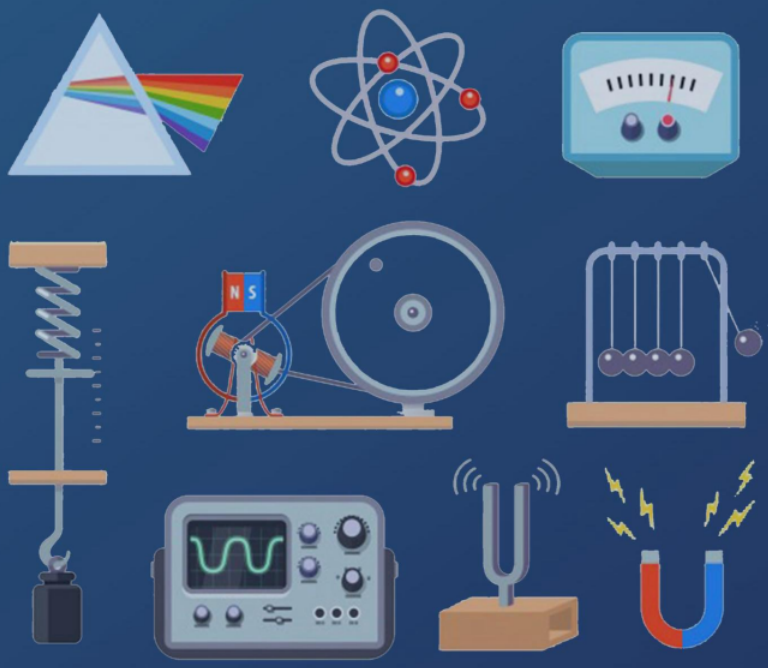
PHYSICS

Magnetism & Matter



SHEET

eSaral.com

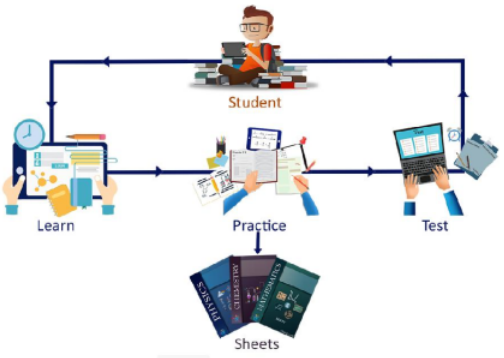
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 2	Page - 3
5.	Answer Key	Page - 5

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-2
1	Introduction to magnetism	1	
2	The bar magnet	2	1
3	The electrostatic analog	3	2,3
4	Bar magnet as an equivalent solenoid	4	
5	Dipole in uniform magnetic field	5-7	4-7
6	Magnetism and gauss's law		
7	Earth's Magnetism	8-10	8-11
8	Why do magnets attract and repel?		
9	Magnetism and magnetic intensity	11-13	12
10	Diamagnetism, paramagnetism and ferromagnetism	14-18	13
11	Hysteresis	19,20	14,15

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Magnetism

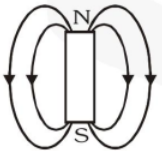
Q1. The geometric length of a bar magnet is 24 cm. The length of the magnet is -

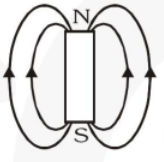
(A) 24cm (B) 28.8cm

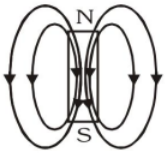
(C) 20cm (D) 30cm

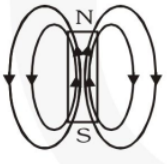
The bar magnet

Q2. The magnetic field lines due to a bar magnet are correct shown in :

(A) 

(B) 

(C) 

(D) 

The electrostatic analog

Q3. The magnetic moment of a short magnet is 8 Am^2 The magnetic induction at a point 20cm away from its mid point on (i) axial point (ii) equatorial point respectively, will be :–

(A) 2×10^{-4} and 10^{-4} T

(B) 3×10^{-4} and 2×10^{-4} T

(C) 4×10^{-4} and 3×10^{-4} T

(D) none of these

Bar magnet as an equivalent solenoid

Q4. A solenoid has 10^3 turns per unit length. On passing a current of 2A, magnetic induction is measured to be 4π Wb/m^2. Calculate magnetic susceptibility of core.

(A) 5000 (B) 4999

(C) 2500 (D) 2499

Dipole in uniform magnetic field

Q5. At two places A and B using vibration magnetometer, a magnet vibrates in a horizontal plane and its respective periodic time 2 sec and 3 sec and at these places the earths horizontal components are B_A and B_B respectively. Then the ratio between B_A and B_B will be :-

(A) 9 : 4 (B) 3 : 2

(C) 4 : 9 (D) 2 : 3

Q6. A magnet of magnetic moment 4A–m^2 is held in a uniform magnetic field 5×10^{-4}T with the magnetic moment vector makes an angle 30° with the field. Work done in increasing the angle from 30° to 45°:-

(A) 3.2×10^{-4} J

(B) 1.6×10^{-4} J

(C) 1.6×10^{-3} J

(D) 3.2×10^{-3} J

Q7. A bar magnet has a magnetic moment equal to 5×10^{-5} A-m^2. It is supended in a magnetic field which has a magnetic induction B equal to $8\pi \times 10^{-4}$ Tesla. The magnet vibrates with a period of vibration equal to 15 sec. The moment of inertia of the magnet is -

(A) 5.2×10^{-7} kg-m^2

(B) 11.25×10^{-7} kg-m^2

(C) 22.5×10^{-7} kg-m^2

(D) 7.16×10^{-7} kg-m^2

Earth's Magnetism

Q8. At a certain place, the horizontal component B_0 and the vertical component V_0 of the earth's magnetic field are equal in magnitude. The total intensity at the place will be :

(A) B_0 (B) B_0^2

(C) $2B_0$ (D) $\sqrt{2}\,B_0$

Q9. At Geo-magnetic poles, the angle of dip is:-

(A) 45° (B) 30°

(C) zero (D) 90°

Q10. The vertical component of earth magnetic field is zero at or the earth's magnetic field always has a vertical component except at the :-

(A) magnetic poles

(B) geographical poles

(C) every place

(D) magnetic equator

Magnetism and magnetic intensity

Q11. Using mass (M), length (L), time (T) and current (A) as fundamental quantities, the dimension of permeability is :-

(A) $[M^{-1}LT^{-2}A]$ (B) $[ML^2T^{-2}A^{-1}]$

(C) $[MLT^{-2}A^{-2}]$ (D) $[MLT^{-1}A^{-1}]$

Q12. The magnetic susceptibility of a para-magnetic substance is 3×10^{-4}. It is placed in a magnetising field of 4×10^4 A/m. The intensity of magnetisation will be :-

(A) 3×10^8 A/m (B) 12×10^8 A/m

(C) 12 A/m (D) 24 A/m

Q13. The magnetic moment of a magnet of mass 75 gm is 9×10^{-7} A-m^2. If the density of the material of magnet is 7.5×10^3 kg/m^3 then intensity of magnetisation will be :-

(A) 0.9 A/m (B) 0.09 A/m

(C) 9 A/m (D) 90 A/m

Diamagnetism, paramagnetism and ferromagnetism

Q14. Relative permittivity and permeability of a material are ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material ?

(A) $\varepsilon_r = 0.5, \mu_r = 1.5$

(B) $\varepsilon_r = 1.5, \mu_r = 0.5$

(C) $\varepsilon_r = 0.5, \mu_r = 0.5$

(D) $\varepsilon_r = 1.5, \mu_r = 1.5$

Q15. The material of permanent magnet has :-

(A) High retentivity, low coercivity

(B) Low retentivity, high coercivity

(C) Low retentivity, low coercivity

(D) High retentivity, high coercivity

Q16. When a diamagnetic substances is inserted in a current carrying coil, then magnetic field is :-

(A) Decreased

(B) Unchanged

(C) Increased

(D) Increased or decreased depending upon the relative volume of the substance

Q17. The value of magnetic susceptibility for super–conductors is :-

(A) Zero (B) Infinity

(C) +1 (D) –1

Q18. If the magnetic susceptibility of a magnetic material is – 0.004 then its nature will be :-

(A) Diamagnetic

(B) Paramagnetic

(C) Ferromagnetic

(D) Non magnetic

Hysteresis

Q19. Which of the following statements is incorrect about hystersis :-

(A) this effect is common to all ferrro-magnetic substances

(B) the hysteresis loop area is proportional to the thermal energy developed per unit volume of the material

(C) the hysteresis loop area is independent of the thermal energy developed per unit volume of the material

(D) the shape of the hysteresis loop is characterstic of the material

Q20. Soft iron is used to make the core of transformer, because of its :

(A) low coercivity and low retentivity

(B) low coercivity and high retentivity

(C) high coercivity and high retentivity

(D) high coercivity and low retentivity

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The magnetic lines of force inside a bar magnet:

(A) are from north-pole to south-pole of the magnet

(B) do not exist

(C) depend upon the area of cross-section of the bar magnet

(D) are from south-pole to north-pole of the magnet

Q2. A bar magnet is held perpendicular to a uniform field. If the couple acting on the magnet is to be halved by rotating it, the angle by which it is to be rotated is :-

(A) 30° (B) 60° (C) 45° (D) 90°

Q3. The magnetic field at a point X on the axis of a small bar magnet is equal to the field at a point Y on the equator of the same magnet. The ratio of the distances of point X and Y from the centre of the magnet is:-

(A) 2^{-3} (B) $2^{-1/3}$ (C) 2^{3} (D) $2^{1/3}$

Q4. Magnetic moments of two identical bar magnets are M and 2M respectively. Both are combined in such a way that their simmilar poles are together. The time period in this situation is T_1. If polrity of one of the magnet is reversed then its period becomes T_2 then :-

(A) $T_1 > T_2$

(B) $T_1 < T_2$

(C) $T_1 = T_2$

(D) None of these

Q5. A magnetic needle is kept in a non-uniform magnetic field. It experiences-

(A) a torque but not a force

(B) neither a force nor a torque

(C) a force and a torque

(D) a force but not a torque

Q6. The time period of oscillation of a magnet in vibration magnetometer is 1.5 second. The time period of oscillation of another magnet similar in size, shape and mass but having one fourth magnetic moment that of the same place will be :-

(A) 0.75 s (B) 1.5s

(C) 3.0 s (D) 6.0 s

Q7. A magnetic needle is made to vibrate in uniform field B_H, then its time period is T. If it vibrates in the field of intensity $4B_H$, it time period will be :-

(A) 2T (B) $\frac{T}{2}$

(C) $\frac{2}{T}$ (D) T

Q8. The horizontal component of earth's magnetic field at place is $0.36x10^{-4}$ weber/m^2 . If the angle of dip at that place is 60° then the value of vertical component of earth's magnetic field will be (in wb/m^2)

(A) $6x10^{-5}T$ (B) $6\sqrt{2}x10^{-5}T$

(C) $3.6\sqrt{3}\times10^{-5}T$ (D) $\sqrt{2}x10^{-5}T$

Q9. If the angle of dip at two places are 30° and 45° respectively, then the ratio of horizontal component of earth's magnetic field at two places will be :-

(A) $\sqrt{3}:\sqrt{2}$ (B) $1:\sqrt{2}$

(C) $1:\sqrt{3}$ (D) 1 : 2

Q10. A magnetic needle suspended horizontally by an unspun silk fibre, Oscillates in the horizontal plane because of the restoring torque originating mainly from :-

(A) the torsion of the silk fibre

(B) the force of gravity

(C) the horizontal component of earth's magnetic field

(D) all the above factors

Q11. The value of earth magnetic field $B_H = 0.3$ gauss. In this magnetic field a magnet is oscillating with 5 oscillation/min. To increase the oscillation of magnet upto 10 oscillation/min. The value of earth magnetic field increased by :-

(A) 0.3 gauss (B) 0.6 gauss

(C) 0.9 gauss (D) 0.12 gauss

Q12. If the magnet of length 10 cm and pole strength 40 Am is placed at an angle of 45° in an uniform induction field of intensity 2×10^{-4} tesla. The couple acting on it will be :-

(A) 0.656×10^{-5} N–m

(B) 0.656×10^{-4} N–m

(C) 0.5656×10^{-4} N–m

(D) 0.5656×10^{-3} N–m

Q13. Which of the following statements is correct for ferromagnetic material :-

(A) These become diamagnetic at Curie temperature

(B) These become paramagnetic at Curie temperature

(C) Their magnetic susceptibility becomes zero at Curie temperature

(D) Its magnetic perperties are explained on the basis of electron principle

Q14. The mass of a specimen of a ferromagnetic material is 0.6 kg. and its density is 7.8×10^3 kg/m^3. If the area of hysteresis loop of alternating magnetising field of frequency 50Hz is 0.722 MKS units then the hysteresis loss per second will be :-

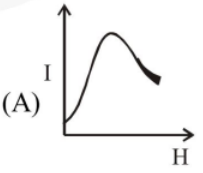
(A) 277.7×10^{-5} J/sec.

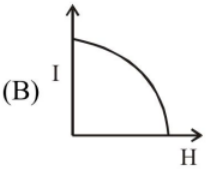
(B) 277.7×10^{-6} J/sec.

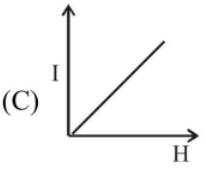
(C) 277.7×10^{-4} J/sec.

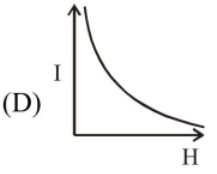
(D) 277.7×10^{-3} J/sec.

Q15. The correct I–H curve for paramagnetic materials is :-

(A) 

(B) 

(C) 

(D) 

Answer Key

Ex - 1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	D	A	B	A	A	D	D	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	B	B	D	A	D	A	C	A

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	D	B	C	C	B	C	A	C
11	**12**	**13**	**14**	**15**					
C	D	B	A	C					

-: Important Notes & Formulas :-